coolant. One of the big challenges for Nair and Balan was to ensure a uniform cooling passage. They achieved it with such great precision that the fourth stage has been a success consistently since the second flight of PSLV in 1994.

The commitment of virtually every team member was inspiring in itself. Such was the dedication of Kuppuraj, our engineer, who worked on the turbo pump that even while battling a terminal disease, he insisted on coming to work. When I tried to advise him against working, he said, 'Sir, have you already written me off?' He did not live to see the PSLV fly.

Vikas was a bigger engine with greater complexities. Here we were, having put together the engine and ground tested it, asked to do it again for a real flight of India's first Polar Satellite Launch Vehicle (PSLV). And this meant much more than reassembling the engine and mounting it on the rocket. An engine was the core of a stage (and PSLV had four stages), but there were many other components that constituted a stage. There were the tankages that supplied the fuel, oxidiser and water to the engine; the gas bottles that held the gas to pressurise the tankages; and complicated fluid circuits, valves and regulators.

A maze of cables and pipelines had to interconnect several parts of the engine. Some had to be fitted into interstages, the buffer between two stages of a rocket; others had to be systematically packed and into the cylinder that is the body of the rocket. We were waking up to a challenge we knew from the day we landed in Vernon in 1974. The contract covered only the making of the engine that included the thrust chamber, the gas generator and the turbo pump. All the rest of the stage—tankages, inter-stage structures, thrust frame, fill-and-drain systems, pressurisation systems, pogo and splash, overflow systems, gymbal activation system and much more—we had to develop on our own. Without these subsystems a flying rocket would remain a grounded dream.

I was reminded of a golden opportunity we wasted in getting the scope of the contract expanded to cover the stage technology. Dhawan, the man of impeccable integrity, had spoiled it. The year was 1975, and Airbus, the